



SIMATS ENGINEERING

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Course Code: CSA1206

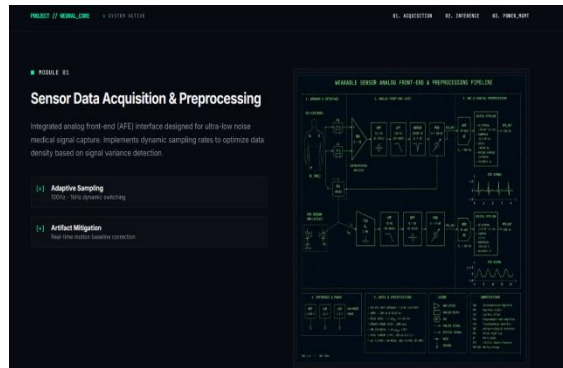
Slot: D

Course Name: Computer Architecture

Course Faculty: Dr. Vasavi.J

Project Title: Neural Network Accelerator for Wearable IOT Health Monitoring

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)



Project Description: (here you write what you did in this project (contribution) including Model Description)

The Neural Network Accelerator for Wearable IoT Health Monitoring is designed to provide low-power and real-time processing of physiological signals such as ECG, PPG, and motion data. The system collects and preprocesses sensor data by removing noise, normalizing signals, and extracting important features. A lightweight neural network then analyzes the processed data to detect health patterns and generate predictions or alerts. Hardware optimization techniques such as quantization, sparsity, and near-sensor computation are used to reduce computation, memory usage, and energy consumption, making the system suitable for resource-constrained wearable devices. The accelerator also integrates closely with the wearable processor and sensors to ensure smooth data flow and fast processing. It activates the required hardware only when necessary, helping to conserve battery power while maintaining stable operation. This approach enables efficient, accurate, and reliable on-device health monitoring, making the system suitable for continuous use in different wearable IoT applications.

Student Signature

Guide Signature